MIL-D-52277A(ME)
24 June 1976
SUPERSEDING
MIL-D-52277(MO)
15 March 1963

MILITARY SPECIFICATION

DRIER, INFRARED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an infrared drier for use in a preservation and packing set assigned National Stock Number (NSN) 3540-00-293-9180.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.

FSC 4440

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MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible, and Extra-Flexible, 300 and 600 Volts).

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 101	- Preservation, Packaging, and Packing Materials: Test Procedures.
FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel.
A 569 - Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The infrared drier (hereinafter referred to as "drier") shall be as shown on Figures 1 through 4 and as specified herein. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 First article (first-produced drier). The contractor shall furnish one or more driers as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce driers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Steel plates, shapes, and bars. The steel plates, shapes, and bars shall conform to ASTM A 36.

3.3.2 Steel sheet. The steel sheet shall conform to ASTM A 569, condition and finish in accordance with the contractor's standard practice.

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3.3.3 Bolts. Bolts shall conform to FF-B-575, Type 2, Class 1.

3.3.4 Nuts. Nuts shall conform to FF-N-836, Type II, Style 4.

3.4 Safety. Electrical circuits shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

3.5 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, replacement, and service in the field without the removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to the driers.

3.6 Construction. The drier shall be complete as to mechanical and electrical construction and shall withstand the shock and vibration encountered when mounted in a van-type trailer traversing rough terrain. General construction of the drier heating element bank housings shall be as shown on Figures 1 through 4. Free air space shall be provided within the housings to provide for cool operation of the heating element bases, sockets, and wiring system. The rear, bottom, and top of the housings shall be perforated with 1/4-inch x 4-inch slots or 1/2-inch-diameter holes, symmetrically placed in a pattern that will prevent entrapment of combustible gases in the free air spaces. If separators are used horizontally between the individual rows of heating elements within the housings, the separators shall be perforated to allow free movement of air in the vertical plane. The wheels for the movable bank of the drier shall operate freely without binding. The movable bank of the drier shall incorporate a locking or clamping device to prevent movement of the drier bank when the drier is in use or the trailer is in motion. Heating elements shall be provided with guards to prevent damage by items being dried.

3.7 Conveyor. The conveyor rack shall be a standard commercial item mounted as shown on Figures 1 through 4.

3.8 Electrical system. Each bank assembly or side of the drier shall operate independently of the other. Each bank shall be designed to operate on a 208-volt, 3-phase, 4-wire, 50/60-Hz power supply, with individual rows of heating elements to operate on 120 volts. The load shall be balanced between phases. Each bank shall be equipped with a 10-foot, rubber-covered, 4-wire, bare-ended cable conforming to MIL-C-3432, Light Duty, Oil Resistant, for connection to the power supply. Total power consumption shall be not more than 18 kw, plus or minus 5 percent.

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3.8.1 Heating elements. The drier shall be equipped with 48 tungsten-filament, 375-watt, 115-volt, R-40, 1F bulb, medium 5C base, infrared lamps, or an equivalent number of far-infrared tubular lamps, as specified (see 6.2). The surface temperature of the far-infrared tubular lamps shall not exceed 465° F.

3.9 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.9.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.9.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.10 Identification marking. The drier shall be identified in accordance with MIL-STD-130. The marking shall be applied to the drier on plates conforming to MIL-P-514, Type I, Style 3, Composition C of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.11 Instruction plates. Each drier shall be equipped with instruction plates or diagrams, including warnings and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the drier. Instruction plates shall conform to MIL-P-514, Type III,

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Composition C of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.12 Treatment and painting. The portions of the drier normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2), except that electrical components and parts subject to high temperatures shall be treated in accordance with the contractor's standard practice.

3.13 Workmanship. The quality of workmanship shall be such as to produce driers that are in accordance with the requirements of this specification and are constructed so as to insure proper functioning of all parts of the unit.

3.13.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cuase injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.13.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the driers to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the driers shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced drier inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced drier inspection.

4.3.1 Examination. The first-produced drier shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced drier shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level S2.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The drier shall be examined as specified herein for the following defects:

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Major

101. Dimensions or tolerances not as specified.
102. Materials not as specified.
103. Safety provisions not as specified.
104. Ease of maintenance provisions not as specified.
105. Construction not as specified.
106. Conveyor not as specified.
107. Fungus-and-moisture-resistant provisions not as specified.
108. Instruction plates not as specified.
109. Workmanship not as specified.

Minor

201. Identification marking incorrect, illegible, or missing.
202. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Operational. The drier shall be connected to the power source specified in 3.8 and energized for not less than 5 minutes. Total power consumption shall be measured with a watt hourmeter. If far-infrared tubes are used, the surface temperature of the tubes shall be determined with a thermocouple. Failure of any heating element to operate, total power consumption of more than 18 kw, plus or minus 5 percent, a surface temperature of a tube, if used, of more than 465° F, or the load not balanced between phases shall constitute failure of this test.

4.5.2.2 Mobility. The drier in its normal operating condition shall be secured within the body of a van-type truck. The drier shall then be transported a distance of 50 miles over unimproved roads at the highest permissible speed. Permissible speed is defined as the highest safe speed for this type of vehicle in the area where the test is being conducted, taking into account road alinement, surface conditions, and weather. The road surface shall include one or more of the following types of imperfections ranging in depth or height of 2 inches to 8 inches below or above the surface of the road: bumps, washboard effect, potholes, ditches, or broken irregular areas. The surface shall be capable of twisting, jarring, shaking, or vibrating the drier components. During the test, the vehicle shall be maneuvered to contact the surface imperfections as often and as continuous as practicable. Upon completion of the mobility test, the drier shall be tested as specified in 4.5.2.1.

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Failure of any heating element to operate, cracking or breaking of any weld, damaging or loosening of any component, loosening or breaking of any bolted or riveted connections, inability to open or close the cabinet doors, or malfunction of locking or clamping device shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The preproduction pack of the crated drier shall be subjected to one of the following tests. The railroad car impact test shall be used when such facilities are available. The pendulum impact test is an alternate test. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, blocking, or bracing, or any visible damage to the contents shall constitute failure of the test and shall be cause for rejection.

4.6.1.2.1 Railroad car impact. The crated drier shall be subjected to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set.

4.6.1.2.2 Pendulum impact. The crated drier shall be subjected to the pendulum impact test in accordance with FED. TEST METHOD STD. No. 101, Test Method 5012.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

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110. Materials or container not as specified.
111. Heating elements not removed, unit packed, or secured as specified.
112. Sockets or any reflecting surfaces not covered as specified.
113. Cable not coiled and secured as specified.
114. Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the preproduction model or a production drier. When the preproduction model is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production driers.

5.2 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.2.1 Level A. The heating elements shall be removed. The sockets and any reflecting surfaces of the drier shall be covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The barrier material shall be secured in place with tape conforming to PPP-T-60, Type IV. The removed heating elements shall be unit packed in commercial containers and shall be placed and secured in the drier cabinet. The cable shall be coiled and secured to the drier.

5.2.2 Commercial. Driers shall be preserved in accordance with FED. STD. No. 356.

5.3 Packing. Packing shall be Level A or Commercial, as specified (see 6.2).

5.3.1 Level A. Each complete drier, preserved as specified in 5.2, shall be packed in a crate conforming to MIL-C-104, Type I, Class 2, Style a. Blocking, bracing, and anchoring shall be in accordance with the appendix to the crate specification and MIL-STD-1186.

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5.3.2 Commercial. Driers shall be packed in accordance with FED. STD. No. 356.

5.4 Marking.

5.4.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.4.2 Commercial packaging. Marking shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The driers are used for drying items being cleaned in a preservation operation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first-produced drier and number of driers required (see 3.2).
(c) Type of heating elements required (see 3.8.1).
(d) Color required (see 3.12).
(e) When welding process will not be at the option of the contractor (see 3.13.2).
(f) Time frame required for submission of preproduction pack (see 5.1).
(g) Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First-produced drier. Any changes or deviations of production driers from the approved first-produced drier during production will be subject to approval of the contracting officer. Approval of the first-produced drier will not relieve the contractor of his obligation to furnish driers conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the driers in accordance with this specification.

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6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each drier.

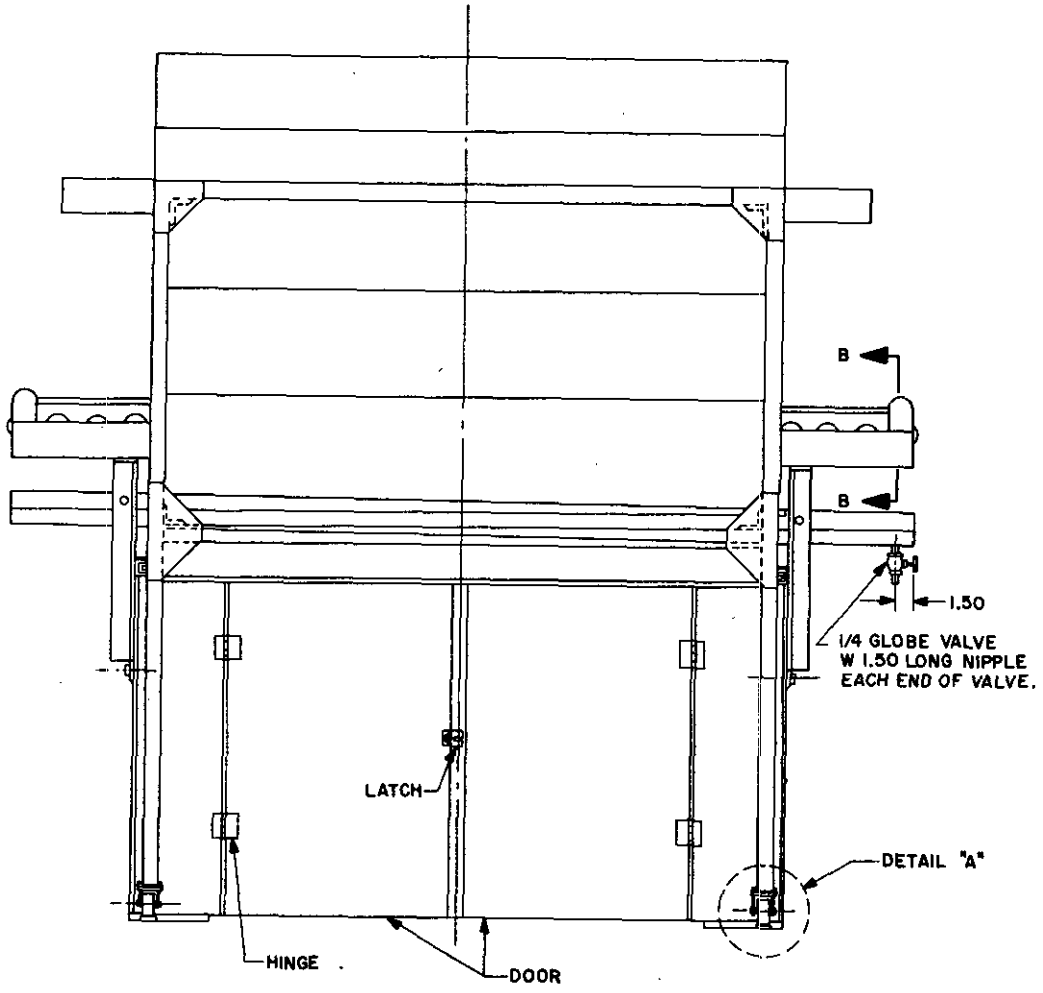
6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of driers.

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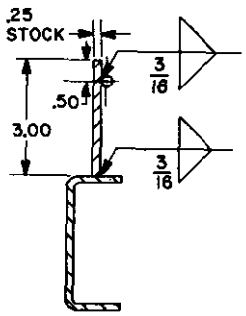
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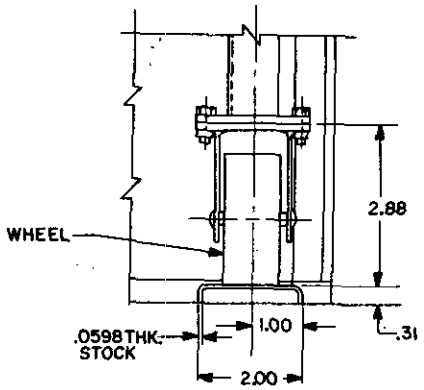
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FRONT ELEVATION



SECTION B-B



DETAIL "A"

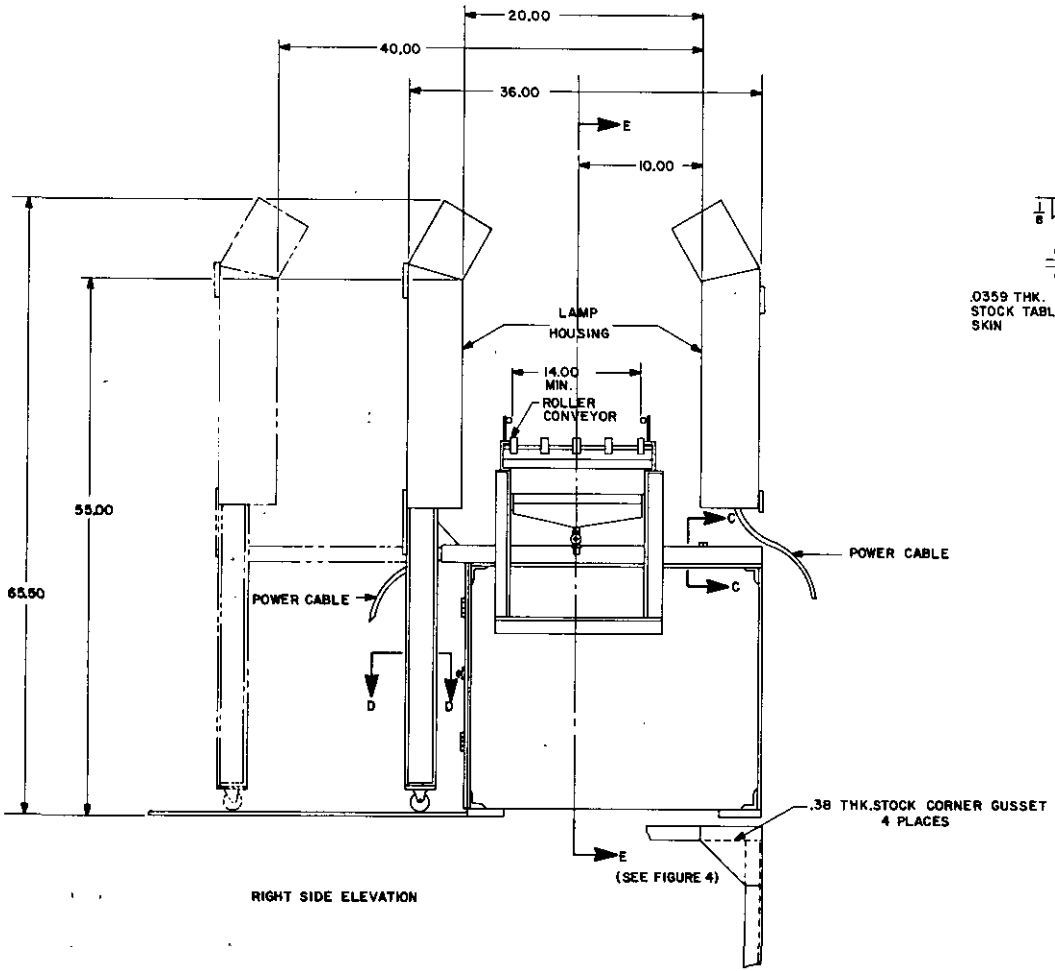
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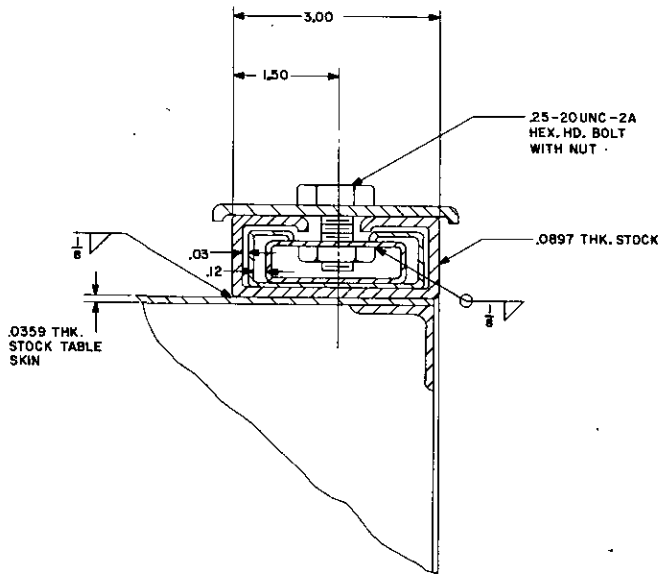
1. ALL WELDED STEEL CONSTRUCTION UNLESS OTHERWISE NOTED.
2. WELD SYMBOLS ARE IN ACCORDANCE WITH AWS A2.0.
3. ALL DIMENSIONS ARE IN INCHES.
4. UNSPECIFIED TOLERANCE ± .03.
5. NOTES APPLY TO ADDITIONAL VIEWS ON FIGURES 2, 3, AND 4.

FIGURE 1. DRIER, INFRARED

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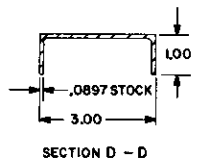


FIGURE 2. DRIER, INFRARED

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